

Data Warehousing for the JMA Wireless Dome

DataCrafters

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Project Goals

- Design an easily implementable data warehouse to allow for efficient storage and BI solutions.
- Add value by providing insights about concession and merchandise sales, promotions, and ticket allocations.





Project Plan

- Create a high-level dimensional model (bus matrix).
- Read data into Snowflake in CSV format.
- Perform ELT Processes within Snowflake to create dimension and fact tables.
- Connect our Snowflake database to Tableau to provide business intelligence insights.

Data Overview

- We have 10 tables of data: Concessions, Daily Inventory Levels, Games, Merchandise, Products, Promotions, Sales, Sponsors, Suppliers, and Teams.
- From these tables, we created dimension and fact tables that address our business processes.



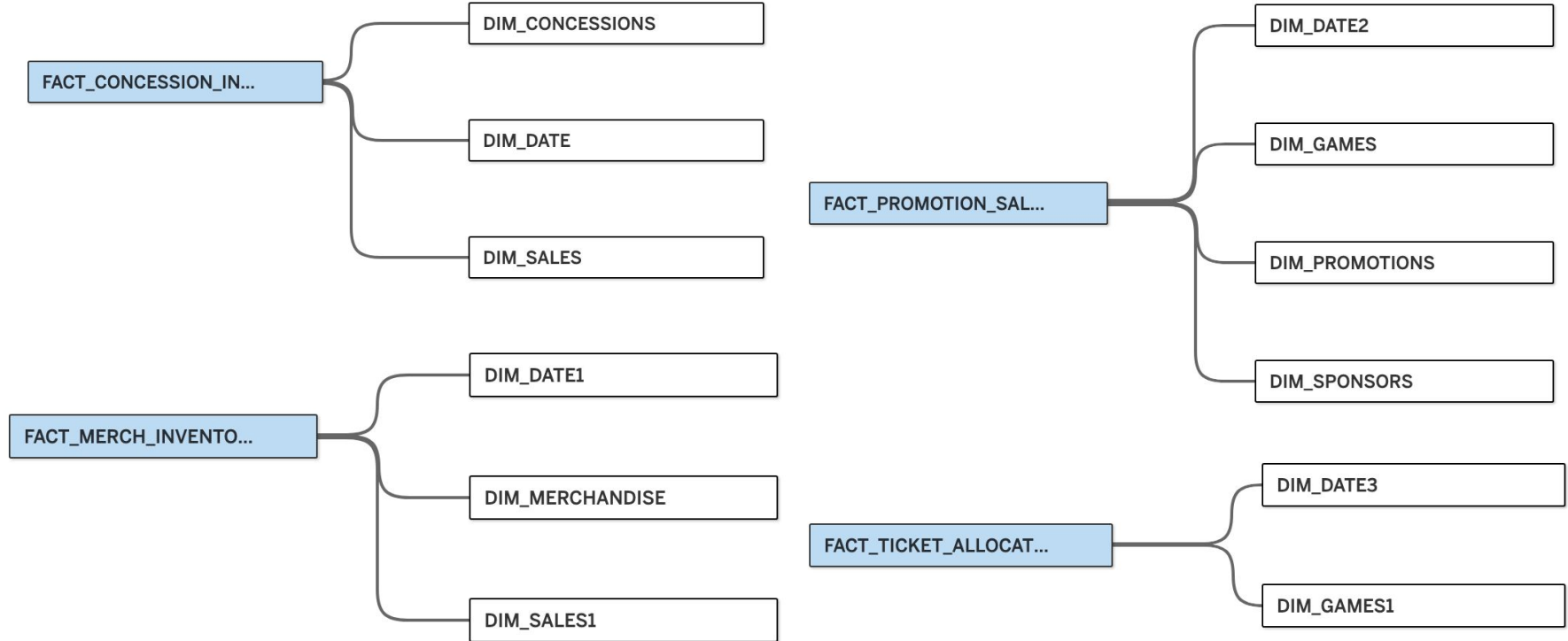
Business Processes

Name of Process	Description
Merchandise Inventory	A snapshot of the amount of stock and units on order for each individual merchandise type following a game.
Concession Inventory	Similar to Merchandise Inventory, this provides the inventory levels after a game, but for concessions.
Promotion Sales	Indicates which sponsors are purchasing promotions. It provides a look at the length of the contract (in games), the cost per game, and the total amount paid.
Ticket Allocations	Details the number of tickets allocated to the away fans for each game, and how many of those tickets get used.

Bus Matrix

Process	Fact Type	Merch	Concess.	Games	Sponsors	Suppliers	Promos	Sales	Date	Facts
Merch Inventory	Periodic Snapshot	✓				✓		✓	✓	Inventory Level, Reorder Date, Reorder Quantity
Concession Inventory	Periodic Snapshot		✓			✓		✓	✓	Inventory Level, Reorder Date, Reorder Quantity
Promotion Sales	Transaction				✓		✓		✓	Contract Length, Total Payment
Ticket Allocations	Transaction			✓				✓	✓	Sport, Tickets Allocated, Tickets Sold, Tickets Unsold

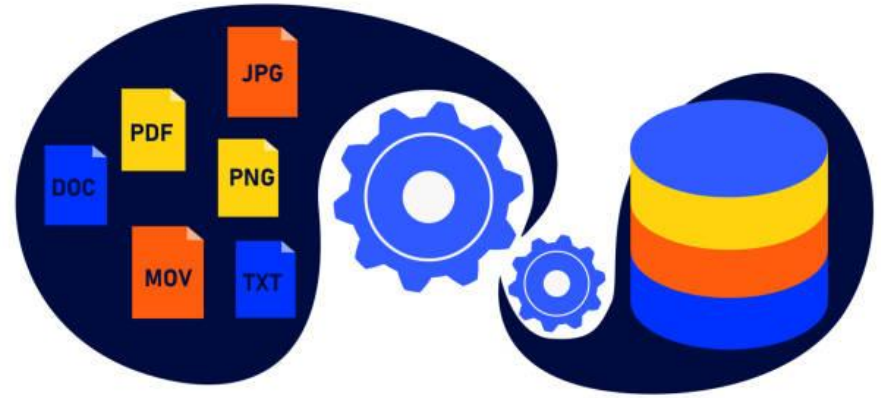
Star Schema



ELT Implementation

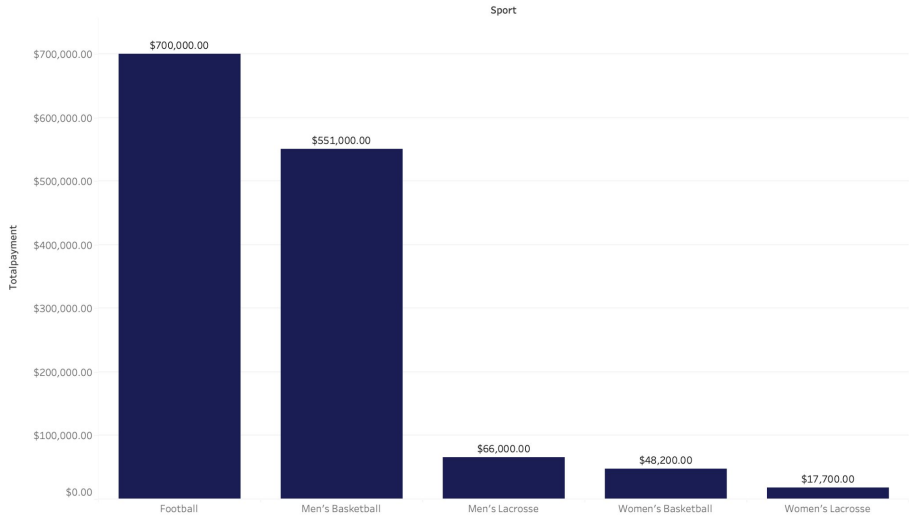
Our ELT process involves the following:

- Extract: Uploaded data from CSVs into our Snowflake stage.
- Load: Loaded the data from the stage into our database environment.
- Transform: Created dimension and fact tables from the data.

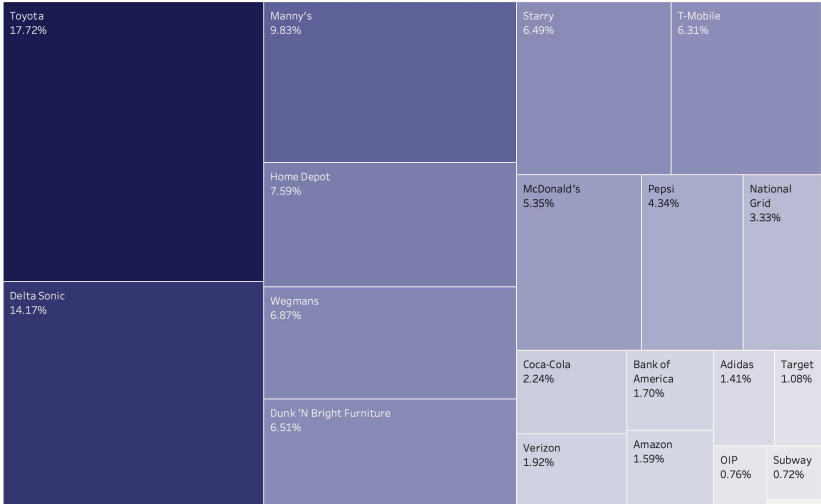


Business Intelligence Example

Promotions Revenue By Sport



Proportion of Promotions Revenue by Sponsor



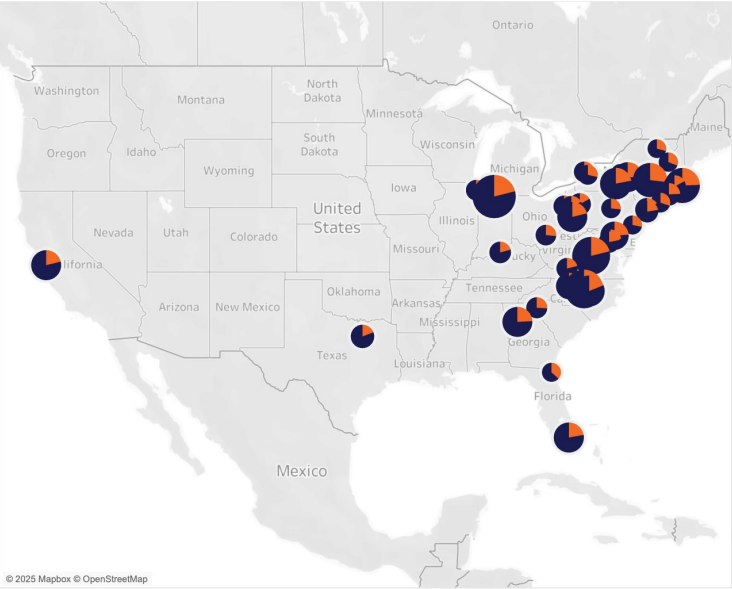
Business Intelligence Demo

SALES OVERVIEW FOR SUPPLIER:

(All)

Sales by Opponent

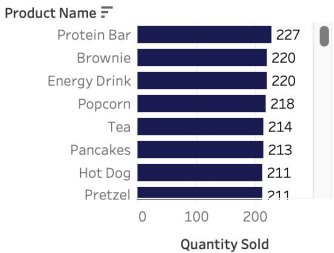
Item Type (All)



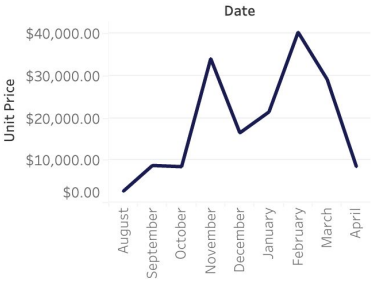
Region

(All)

Items Sold By Category



Total Sales By Month



Challenges and Future Scope

- Challenges

- Lack of publicly available data for the JMA Dome
- Range of event types the Dome hosts

- Future Scope

- Showing off our dashboards to sponsors to gauge their current engagement level
- Planning more efficient uses of sponsorship funds to increase fan engagement
- Potential expansion of merch and concession types

- Conclusion

- We were able to create an expansive data warehouse that contains all the requisite data and information for JMA Dome sales, promotions, and inventory. This will make it easier to store future data as the Dome expands to house other events, and partner with other brands.



An aerial night photograph of the London Stadium, featuring a translucent, illuminated dome with a complex steel truss structure. The stadium is surrounded by a city skyline with numerous lit-up buildings. The text "Thank You!" is overlaid in the center.

Thank You!